

MetaSwarm at AbjadMed: Focal, Class-Balanced Optimization for Diglossic Medical Text in Abjad Scripts

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Problem

Classify Arabic medical text into 82 long-tailed categories. Dialect and unvocalized text complicate the task.

Motivation

Diagnose the tail, then target it.

Per-class F1 guides targeted duplication; linguistic error mechanisms remain hypotheses.

Error structure

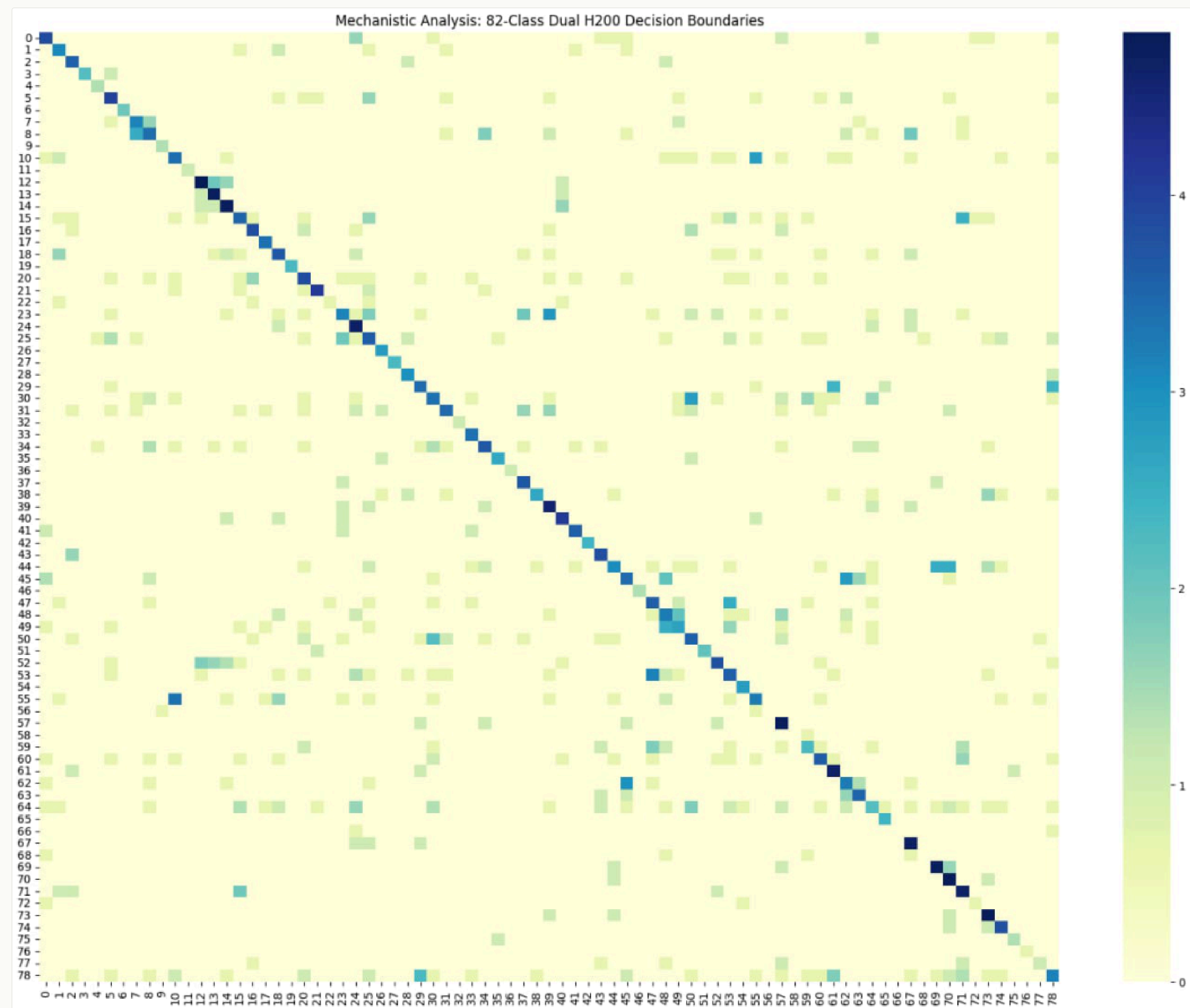


Figure 4. Baseline confusion across the 82 classes.

Method

- 1 Fine-tune CAMELBERT-DA: focal loss + smoothing.
- 2 Duplicate 42 weak classes after each split; average five-fold test logits.

$$L = -\frac{1}{B} \sum_b \alpha_f (1 - p_{t,b})^\gamma \text{CE}_\epsilon(b)$$

Submitted objective, Eq. (2), as printed in the paper.

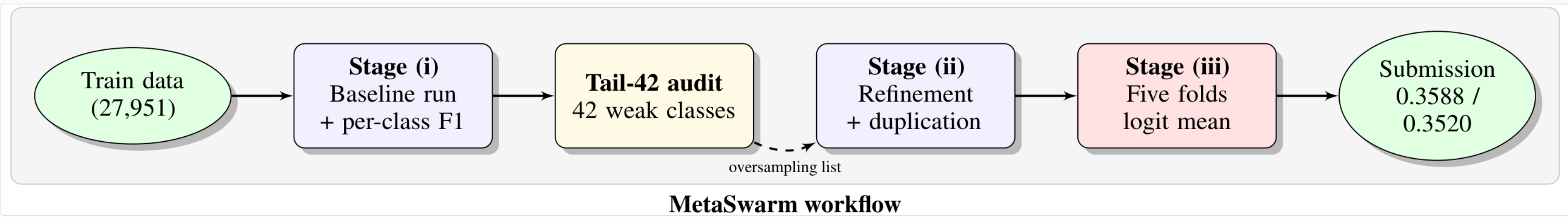


Figure 1. Baseline audit → targeted duplication → five-fold logit mean.
Final loss: $\gamma = 2.5$, $\alpha = 0.25$, $\epsilon = 0.15$. Effective-number weighting is development-only.

Checkpoint selection

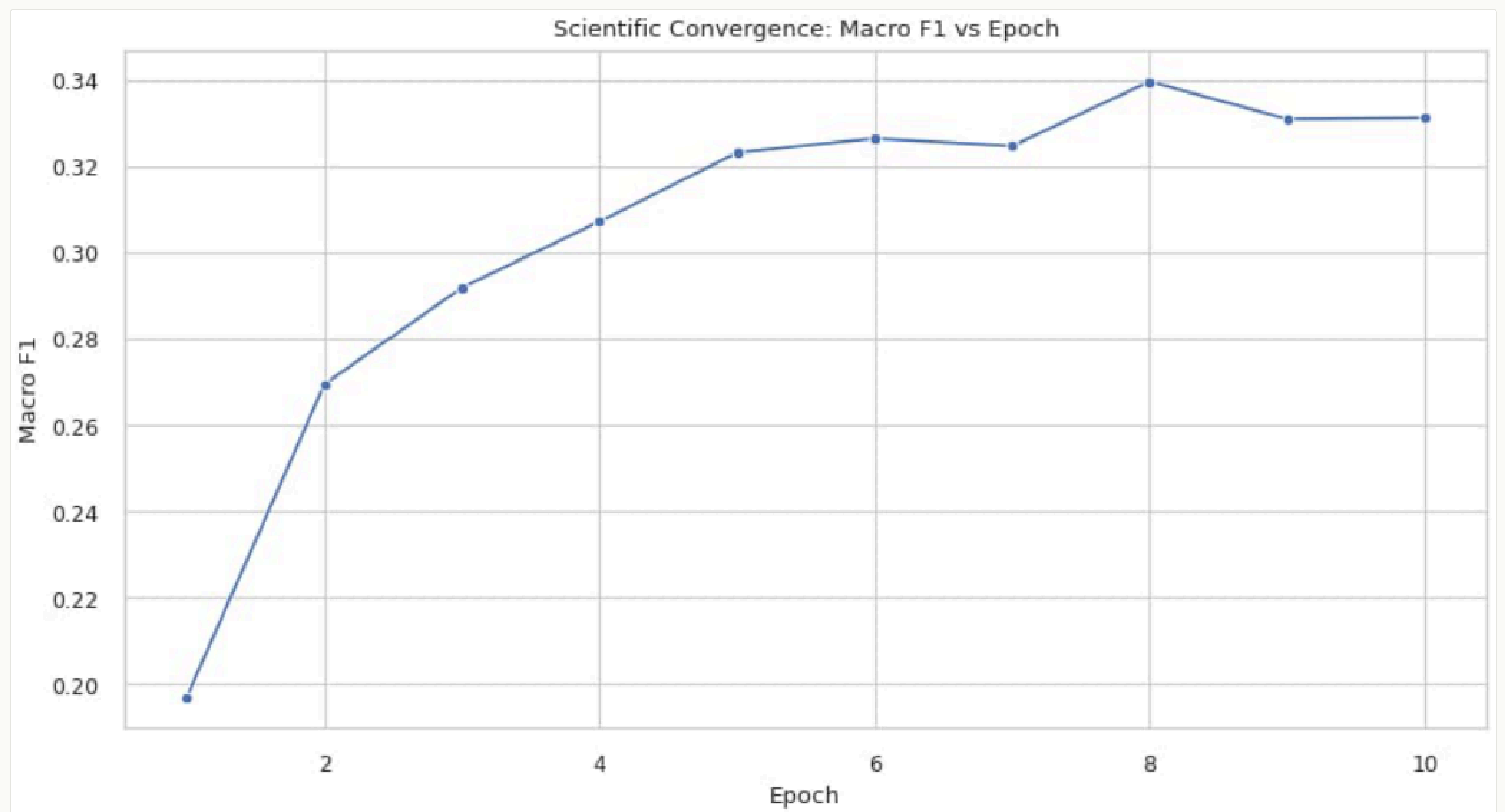


Figure 2. F1 peaks at epoch 8; minimum loss occurs at epoch 4.

Key Results

42/82 classes below baseline F1 = 0.40

SO WHAT → Tail-42 supplies the post-split oversampling list.

Headline Numbers

0.3520

Private macro-F1
Five-fold CAMELBERT-DA ensemble

0.3588 Public macro-F1

42/82 Weak classes

5 Ensemble folds

Tail-42 audit

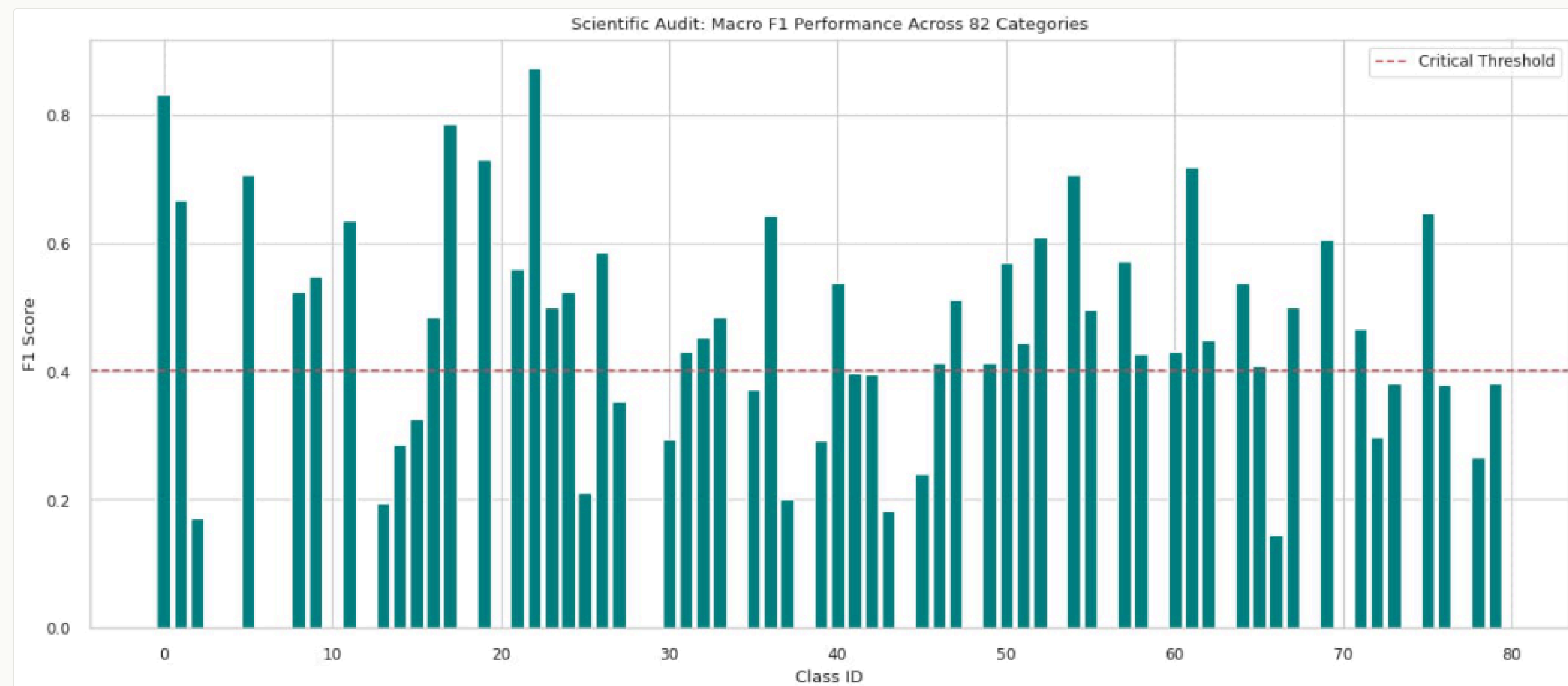


Figure 3. Per-class F1; dashed line marks 0.40.

Takeaway

TAKEAWAY

Tail-42 supplies an intervention list.

No matched ablations establish component gains.